Transformations of reciprocal trigono[redacted]ic functions

1

a $x = \pi n$ where n is an integer.

b $f(x)$ is undefined when $x = \pi n$.

2

a $x = \pi n + \dfrac{\pi}{2}$ where n is an integer.

b $f(x)$ is undefined when $x = \pi n + \dfrac{\pi}{2}$.

3

a $x = \pi n$ where n is an integer.

b $f(x)$ is undefined when $x = \pi n$.

4

a $(0, 1)$

b $\left(\dfrac{\pi}{4}, 1\right)$

c $(0, 7)$

5

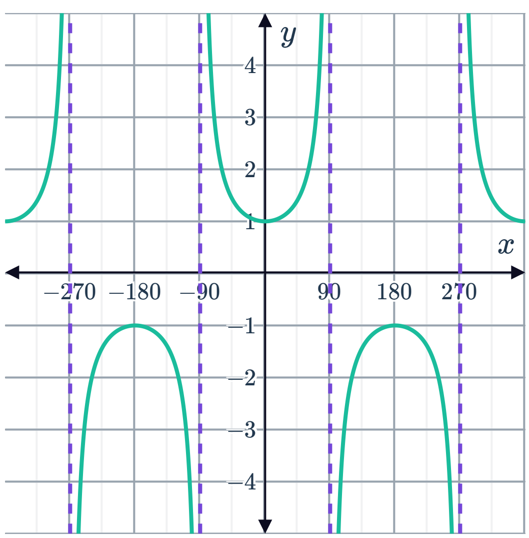
a $\left(\dfrac{\pi}{2}, 5\right)$

b $\left(\dfrac{3\pi}{2}, 5\right)$

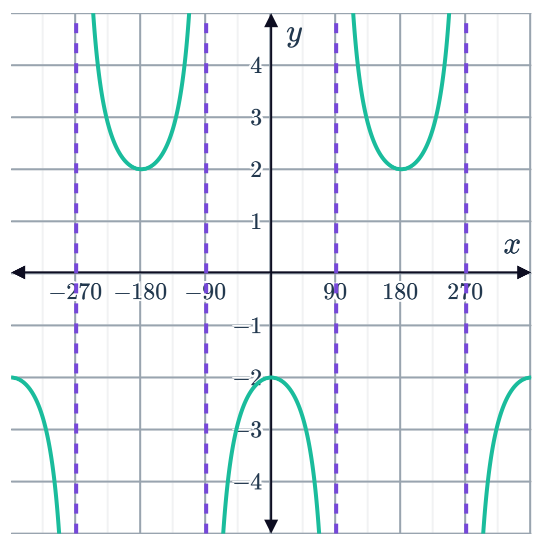
c $\left(\dfrac{\pi}{2}, 3\right)$

6

a

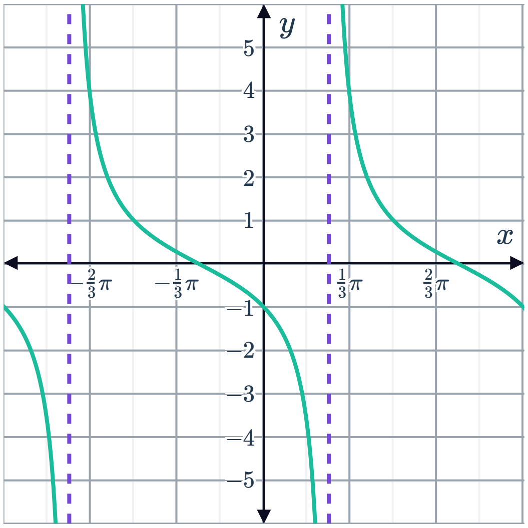


b

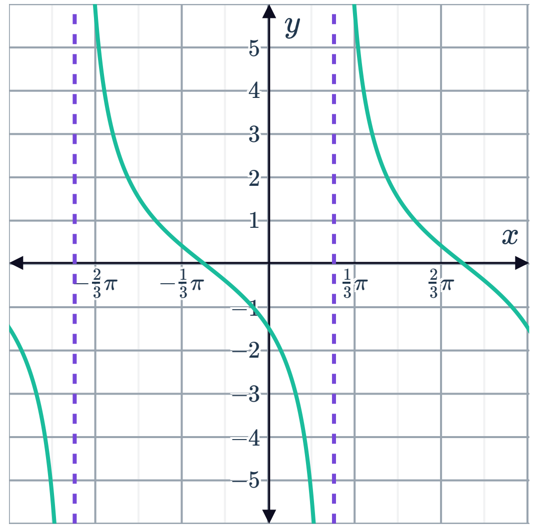


7

a

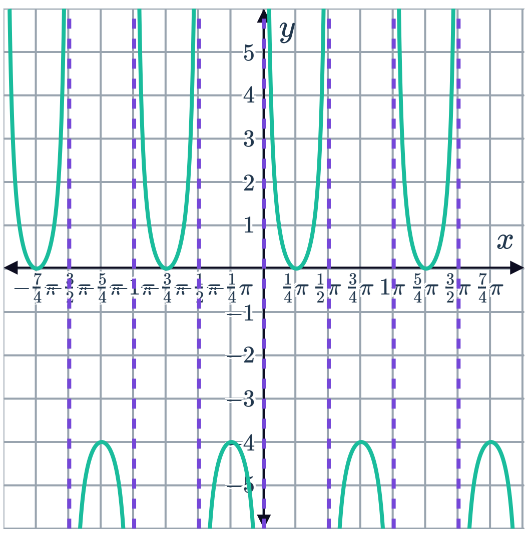

y
x
5
4
3
2
1
−1
−2
−3
−4
−5
$-\frac{2}{3}\pi$
$-\frac{1}{3}\pi$
$\frac{1}{3}\pi$
$\frac{2}{3}\pi$

b

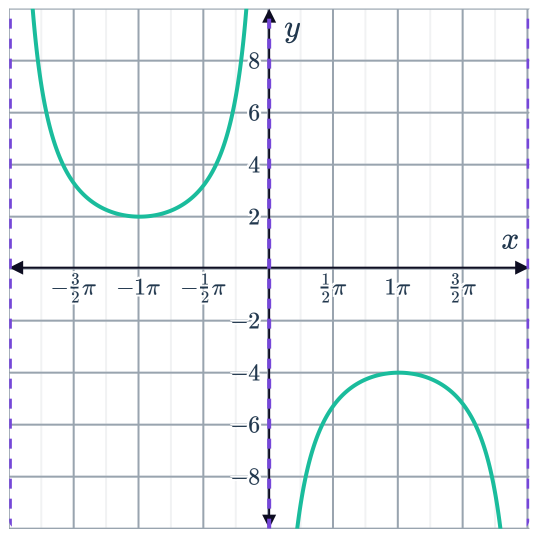

y
x
5
4
3
2
1
−1
−2
−3
−4
−5
$-\frac{2}{3}\pi$
$-\frac{1}{3}\pi$
$\frac{1}{3}\pi$
$\frac{2}{3}\pi$

8

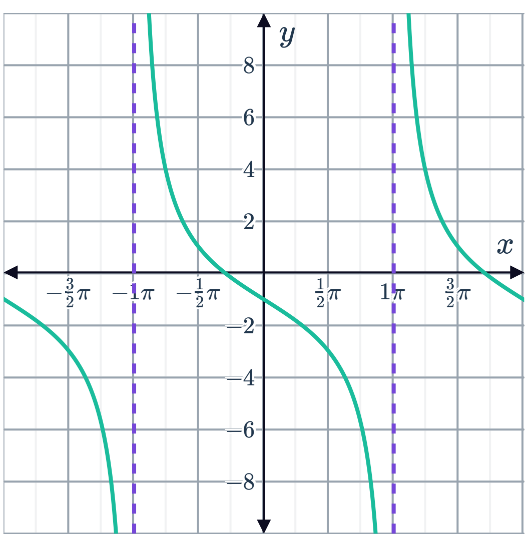
a


y
x
5
4
3
2
1
−1
−2
−3
−4
−5
$-\frac{7}{4}\pi$
$-\frac{3}{2}\pi$
$-\frac{5}{4}\pi$
-1π
$-\frac{3}{4}\pi$
$-\frac{1}{2}\pi$
$-\frac{1}{4}\pi$
$\frac{1}{4}\pi$
$\frac{1}{2}\pi$
$\frac{3}{4}\pi$
1π
$\frac{5}{4}\pi$
$\frac{3}{2}\pi$
$\frac{7}{4}\pi$

b


y
x
8
6
4
2
−2
−4
−6
−8
$-\frac{3}{2}\pi$
-1π
$-\frac{1}{2}\pi$
$\frac{1}{2}\pi$
1π
$\frac{3}{2}\pi$

c

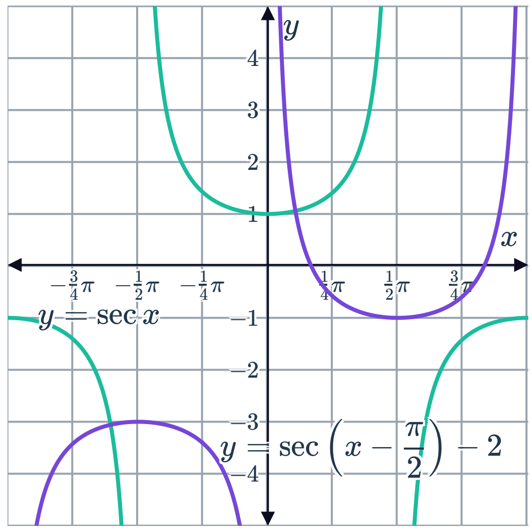

y
x
8
6
4
2
−2
−4
−6
−8
$-\frac{3}{2}\pi$
-1π
$-\frac{1}{2}\pi$
$\frac{1}{2}\pi$
1π
$\frac{3}{2}\pi$

9 Horizontal dilation by a scale factor of $\frac{1}{3}$ then a horizontal translation by $\frac{\pi}{6}$ units to the right.

10

a Move the graph to the right by $\frac{\pi}{2}$ and down by 2 units.

b



c 2π

Domain and range of reciprocal trigonometric functions

11

a Vertical translation 3 units up.

b

i No **ii** Yes **iii** No **iv** No

c $\ldots \cup \left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right) \cup \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{2}\right) \cup \ldots$

d $(-\infty, 2] \cup [4, \infty)$

12

a Vertical translation of 2 units up.

b

i Yes **ii** No **iii** Yes **iv** No

c $\ldots \cup (-2\pi, -\pi) \cup (-\pi, 0) \cup (0, \pi) \cup (\pi, 2\pi) \cup \ldots$

d $(-\infty, 1] \cup [3, \infty)$

13

a Horizontal translation of $\frac{3\pi}{4}$ units to the left.

b

i No **ii** Yes **iii** Yes **iv** Yes

c $\ldots \cup \left(-\frac{7\pi}{4}, -\frac{3\pi}{4}\right) \cup \left(-\frac{3\pi}{4}, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \frac{5\pi}{4}\right) \cup \ldots$

d $(-\infty, -1] \cup [1, \infty)$

14

a Horizontal translation of $\frac{3\pi}{4}$ units to the right.

b

i Yes **ii** Yes **iii** No **iv** No

c $\ldots \cup \left(-\frac{3\pi}{4}, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \frac{5\pi}{4}\right) \cup \left(\frac{5\pi}{4}, \frac{9\pi}{4}\right) \cup \ldots$

d $(-\infty, -1] \cup [1, \infty)$

15

a Horizontal translation of $\frac{3\pi}{4}$ units to the right.

b

i Yes **ii** No **iii** Yes **iv** No

c $\ldots \cup \left(-\frac{5\pi}{4}, -\frac{\pi}{4}\right) \cup \left(-\frac{\pi}{4}, \frac{3\pi}{4}\right) \cup \left(\frac{3\pi}{4}, \frac{7\pi}{4}\right) \cup \ldots$

d $(-\infty, \infty)$

16

a Horizontal compression by a factor of $\frac{1}{4}$.

b

i Yes **ii** No **iii** Yes **iv** Yes

c $\ldots \cup \left(-\frac{\pi}{2}, -\frac{\pi}{4}\right) \cup \left(-\frac{\pi}{4}, 0\right) \cup \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \frac{\pi}{2}\right) \cup \ldots$

d $(-\infty, -1] \cup [1, \infty)$

17

a Horizontal stretch by a factor of 3

b

i Yes ii No iii Yes iv Yes

c $\ldots \cup(-6\pi,-3\pi) \cup(-3\pi, 0) \cup(0,3\pi) \cup(3\pi, 6\pi) \cup \ldots$

d $(-\infty, \infty)$

18

a Horizontal compression by a factor of $\frac{1}{3}$.

b

i Yes ii Yes iii No iv Yes

c $\ldots \cup\left(-\frac{\pi}{2},-\frac{\pi}{6}\right) \cup\left(-\frac{\pi}{6}, \frac{\pi}{6}\right) \cup\left(\frac{\pi}{6}, \frac{\pi}{2}\right) \cup \ldots$

d $(-\infty,-1] \cup[1, \infty)$

19

a Vertical stretch by a factor of 3.

b

i No ii No iii No iv No

c $\ldots \cup(-2\pi,-\pi) \cup(-\pi, 0) \cup(0, \pi) \cup(\pi, 2\pi) \cup \ldots$

d $(-\infty, \infty)$

20

a Vertical stretch by a factor of 4.

b

i No ii No iii Yes iv No

c $\ldots \cup\left(-\frac{3\pi}{2},-\frac{\pi}{2}\right) \cup\left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \cup\left(\frac{\pi}{2}, \frac{3\pi}{2}\right) \cup \ldots$

d $(-\infty,-4] \cup[4, \infty)$

21

a Vertical stretch by a factor of 5.

b

i No ii No iii Yes iv No

c $\ldots \cup(-2\pi,-\pi) \cup(-\pi, 0) \cup(0, \pi) \cup(\pi, 2\pi) \cup \ldots$

d $(-\infty,-5] \cup[5, \infty)$

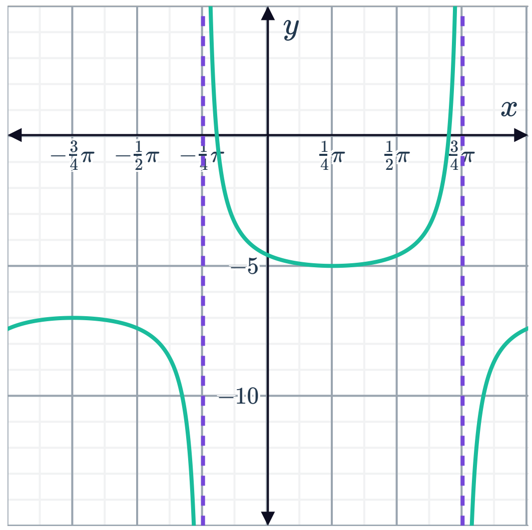
22

a 2π

b $x = \pi n - \dfrac{\pi}{4}$ where n is an integer.

c $\left(\dfrac{\pi}{4}, -5\right), \left(\dfrac{5\pi}{4}, -7\right)$

d



e $\ldots \cup \left(-\dfrac{5\pi}{4}, -\dfrac{\pi}{4}\right) \cup \left(-\dfrac{\pi}{4}, \dfrac{3\pi}{4}\right) \cup \left(\dfrac{3\pi}{4}, \dfrac{7\pi}{4}\right) \cup \ldots$

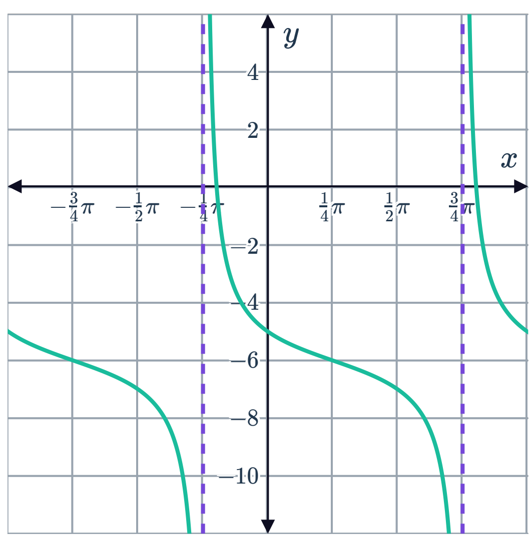
f $(-\infty, -7] \cup [-5, \infty)$

23

a π

b $x = \pi n - \dfrac{\pi}{4}$ where n is an integer.

c



d $\ldots \cup \left(-\dfrac{5\pi}{4}, -\dfrac{\pi}{4}\right) \cup \left(-\dfrac{\pi}{4}, \dfrac{3\pi}{4}\right) \cup \left(\dfrac{3\pi}{4}, \dfrac{7\pi}{4}\right) \cup \ldots$

e $(-\infty, \infty)$

24

a

i 6π

ii 0

iii $y \geq 5, y \leq -5$

b

i $\dfrac{\pi}{3}$

ii 0

iii $y \leq -4, y \geq 4$

c

i 2π

ii $-\dfrac{\pi}{2}$

iii $y \geq 4, y \leq -4$

d

i 2π

ii $\dfrac{\pi}{6}$

iii $y \geq \dfrac{3}{2}, y \leq -\dfrac{3}{2}$

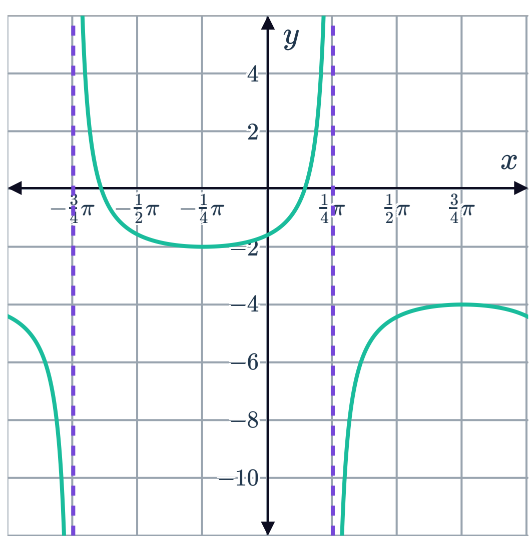
25

a 2π

b $x = \pi n + \dfrac{\pi}{4}$ where n is an integer.

c $\left(\dfrac{7\pi}{4}, -2\right), \left(\dfrac{3\pi}{4}, -4\right)$

d



e $\ldots \cup \left(-\dfrac{7\pi}{4}, -\dfrac{3\pi}{4}\right) \cup \left(-\dfrac{3\pi}{4}, \dfrac{\pi}{4}\right) \cup \left(\dfrac{\pi}{4}, \dfrac{5\pi}{4}\right) \cup \ldots$

Equations of reciprocal trigonometric functions

26

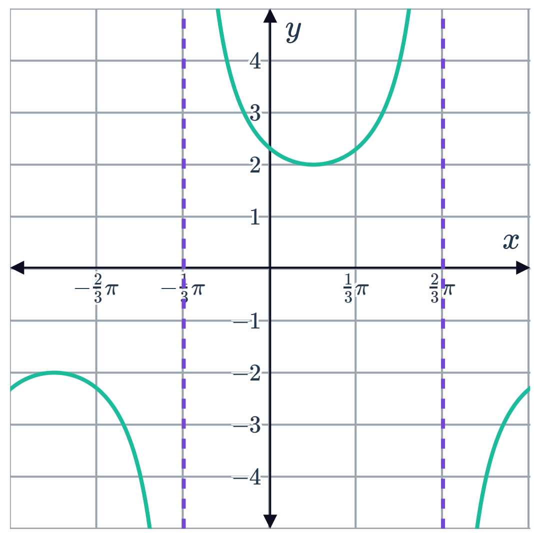
a $y = -\csc x - 3$

b $y = -\sec\left(x + \dfrac{\pi}{2}\right)$

27

a $y = 2\csc\left(x + \dfrac{\pi}{3}\right)$

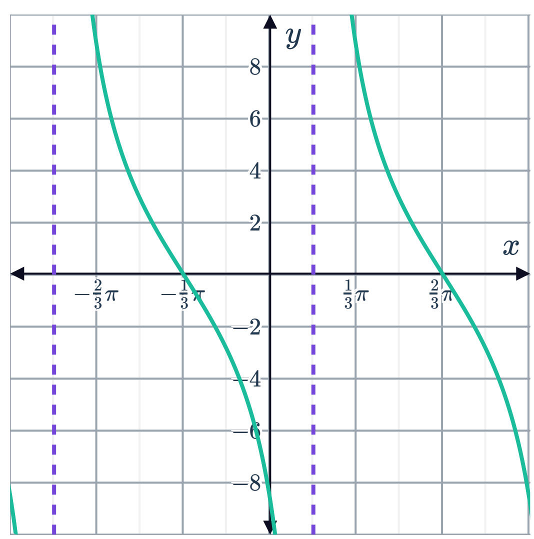
b



28

a $y = 5\cot\left(x - \dfrac{\pi}{6}\right)$

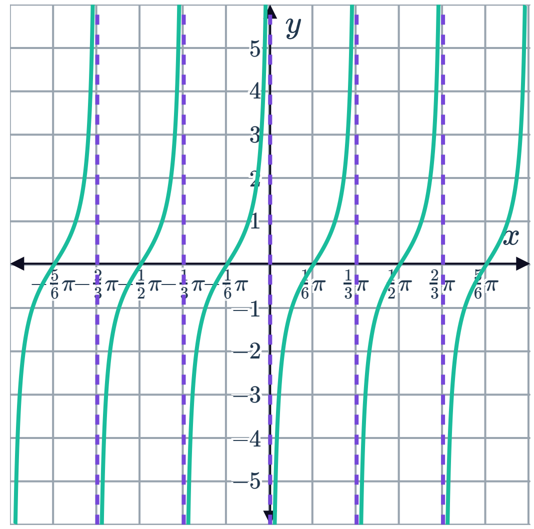
b



29

a $y = -\cot 3x$

b



30

a $x = 0$

b A local minimum.

c 2π

d $y = \csc x$

31

a $x = \dfrac{\pi}{6}$

b A local maximum.

c $\dfrac{2\pi}{3}$

d $y = \sec(3x)$

32

a $x = \pi$

b 2π

c 3

d $y = \csc x + 3$